SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I SORNAVARSHINI.M (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Somavarshini

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	1	1
Accuracy	10%	1	1	1	1
Clarity	5%	1	1	1	1

Total Marks 36 /40

Signature of the Faculty

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests

of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation

Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organizatio n	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professiona l presentatio n	Understanda ble with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CO1 - AT1

SHORT ANSWERS

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COURSE CODE: CSA0109

1. Presentation Layer: Compression and Encryption

The Presentation layer is the sixth layer of OSI model is responsible for data formatting, compression, encryption and decryption. It improves transmission efficiency by reducing data size and enhances security by encrypting the data.

Given data:

Original file size = 900MB

Compression = 40%

Encryption overhead = 5% of the compressed file

1. Calculate the compressed file size:

The file is compressed by 40%.

Compression amount:

$$\frac{40}{100} \times 900 = 360MB$$

Compressed file size:

$$900 - 360 = 540MB$$

2. Calculate the Encrypted File Size

Encryption adds 5% overhead to the compressed file size

Encryption overhead:

$$\frac{5}{100} \times 540 = 27MB$$

Encrypted file size:

$$540 + 27 = 567MB$$

3. Calculate the Total Percentage Reduction:

Reduction in file size:

$$900 - 567 = 333MB$$

Percentage Reduction:

$$\frac{333}{900} \times 100 = 37\%$$

Therefore the final encrypted file size is 567 MB representing a 37% reduction compared to the original file size.

4. How the Presentation layer improves Efficiency and Security.

Compression:

The Presentation Layer reduces data size, saving bandwidth and transmission time.

Encryption:

The Encryption converts readable data into an unreadable format to prevent unauthorised access.

Decryption:

Decryption converts the encrypted data back into readable form at the receiver's end.

Conclusion:

The Presentation Layer improves efficiency and security. The 900 MB file becomes 567 MB after compression and encryption, achieving a 37% size reduction.

2. FTP Data Transfer - Calculation and Explanation:

INTRODUCTION:

The Application layer is the topmost layer of the OSI model. It provides network services to users and applications, such as file transfer, email and web browsing. FTP (File Transfer Protocol) is used to upload and download files over a network.

Given Data:

Total data = 3GB

Request size = 3MB

Throughput = 120Mbps

1. Calculate the total Number of Requests.

Assume 1GB = 1000 MB

Number of requests = $\frac{\text{Total data}}{\text{Request size}}$

$= \frac{3000}{3}$

$= 1000$ requests

2. Calculate the Total Transmission Time

Convert 3GB into megabits:

$3 \times 1000 \times 8 = 24000$ Mb

Transmission time:

$\text{Time} = \frac{\text{Data Size}}{\text{Throughput}}$

$= \frac{24000}{12} = 200$ seconds

Therefore the total transmission time is 200 seconds (3 minutes 20 seconds) assuming constant throughput and ignoring protocol overhead.

3. How the Application Layer Provides Reliable Services.

File Transfer: FTP allows users to upload and download files over a network.

Error Handling: FTP detects transfer problems and reports failures

User authentication: Login credentials help restrict access to authorised users

Data Management: Files are divided into manageable requests for transfer.

Conclusion:

The Application layer provides file transfer services through FTP. For 3GB of data, it generates 1000 segments. and the estimated transmission time is 200 seconds at 120 Mbps.

3. Calculation of End-to-End Delay

Introduction:

The OSI enables communication between devices through different layers. Each layer contributes to the processing delay during data transmission.

Given data:

Application Layer = 4ms

Transport layer = 3ms

Network layer = 5ms

Data link Layer = 2ms

Physical layer = 1ms

Propagation delay = 18ms

Transmission delay = 12ms

1. Total Processing delay:

$4 + 3 + 5 + 2 + 1 = 15ms$

Total Processing delay = 15ms.

2. Total end-to-end delay

Total delay = Processing delay + Propagation delay + Transmission delay

= 15 + 18 + 12

= 45 ms

3. Contribution of each OSI Layer.

Application Layer: Provides network services to applications

Transport Layer: Segments data and supports end-to-end communication.

Network Layer: Selects routes and forwards packets using IP addressing

Data link Layer: Creates frames and performs local link error detection

Physical Layer: Transmits bits as electrical, optical or radio signals.

Propagation delay is the time taken for a signal to travel across the medium. Transmission delay is the time required to place all packet bits onto the link.

Conclusion:

The total end-to-end delay is 45 milli-seconds. Each OSI Layer performs a specific communication function, while transmission and propagation delays contribute to the total time taken for data to travel through the network.

4. Protocol overhead and transmission efficiency

Introduction:

Protocol overhead refers to the extra byts added to frames by OSI layers for addressing and error detection. It increases transmitted data and affects network efficiency

Given:

Useful data = 80MB

Total frame size = 1600 bytes

Protocol overhead per frame = 80 bytes

Assume that the 1600-byte frame size includes the 80-byte overhead.

1. Useful data per frame

Useful data per frame = Frame size - Overhead

= 1600 - 80

= 1520 bytes

2. Calculate the total number of frames:

$$\text{Number of frames} = \frac{\text{Total useful data}}{\text{Useful data per frame}}$$

$$= \frac{80,000,000}{1520}$$

$$= 52631.58 \simeq 52632 \text{ frames.}$$

3. Calculate the total Protocol overhead

Total Overhead = Number of frames × Overhead per frame

= 52632 × 80

= 4,210,560 bytes

Converting bytes to MB:

$= \frac{4210560}{1000000}$

= 4.21 MB (approx)

4. Calculate transmission efficiency:

Efficiency = $\frac{\text{Useful data}}{\text{Total transmitted data}} \times 100$

$= \frac{80,000,000}{80,000,000 + 4,210,560} \times 100 = \frac{80,000,000}{84,210,560} \times 100$

Efficiency ≈ 95%.

5. How does protocol overhead affect network performance?

Bandwidth usage: Overhead consumes additional network bandwidth

Transmission time: Extra bytes increase the time needed to transmit data.

Efficiency: Higher overhead reduces the percentage of useful data transmitted.

Reliable communication: Headers and trailers provide information for addressing, sequencing and error detection

Network Performance: Efficient framing reduces unnecessary overhead and improves bandwidth utilization.

Conclusion:

Approximately 95% of the transmitted data is useful information while the remaining portion is protocol overhead.